Task 2 - ML for Finance. Argimiro Arratia. BSE- 2026

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- Register your team in sheet HW2 of the spreadsheet ML4Finance26_HWteams Sheet HW2 in the drive/Classroom.
- Do not hand in R/Python output; program scripts are hand in separately from the report; write report in LaTeX (preferably) or Word (and convert either to PDF); it is not necessary to keep all decimal digits in reported results.
- Put team members names in the report. Teams should be of at most 3 students (and at least 2).
- Always comment your results (as a Data Scientist would do!).
- **Submit your report in PDF format plus R/Python code and extra data (if any, not given out by me), all in a .zip folder via the Classroom with label: "BSE_Task2-Team - `members names` -". Only one member of team makes the submission for all. Failing to submit as requested will be penalised** -10%

1. (45%) Backtesting trading strategies based on sentiment indicators. Consider the dataset `dataSent.zip` provided in Classroom/Home Work 2 containing the stock price history and associated sentiment indicators for various stocks. Select two and register their tickers in the spreadsheet in column Prob. 1. Try to select different pair from those selected by others (not mandatory). Use the Python (or R equivalent) script `Pylab6_TradingStrategywCost.ipynb` which builds the following compositions of sentiment indicators:

$$\begin{aligned} BULL &= positivePartscr + certaintyPartscr + finupPartscore \\ BEAR &= negativePartscr + uncertaintyPartscr + findownPartscore \\ BBr &= 100 \cdot BULL/(BULL + BEAR) \quad \text{with NAs interpolation} \\ PNlog &= 0.5 \cdot \log((positiveP + 1)/(negativeP + 1)) \end{aligned}$$

and defines performance measures and a trading strategy based on Moving Average crossover ($MAcross$) applied to some explanatory variable (ev). There are two test functions: one for trading the whole period, other for a rolling window analysis over fixed length periods (window-size= 1 yr, 6 mon).

Your team's task is to do the following with your selected stocks

- Design ONE trading strategy (i.e. the signalling sequence) **different** from the $MAcross$ and RSI strategies shown in Lecture 6[1], using any (or all) sentimental indicators
  $ev \in \{positiveP,\ negativeP,\ BULL,\ BEAR, BBr, PNlog\}$, and RVT (volume of news), and test performance of your strategy for the cases where you either allow long-only or long-short trading, and using the rolling window analysis with window sizes = 254 (1 year) or 127 (6 months). Any other parameters of your strategy must be tuned. **In all cases set trading costs to 0.01 (1%).**
  (Suggestions: adapt some of the many strategies from Technical Analysis to the sentimental indicators ev.
  See `https://www.avatrade.com/education/technical-analysis-indicators-strategies`; other examples in `https://www.tradingview.com`, or `https://rpubs.com/johnakwei/207852`, or explore the web for *technical analysis trading strategies*)

In all cases report successful results (where excess return $Me - BH$ is positive); include all metrics of performance seen in class and number of trades done by your strategy. (As an extra try to include trading costs for the rolling window analysis.)

2. (45%) Machine Learning for Option Valuation:

   Use machine learning methods Gaussian Process Regression (GPR) and Neural Networks (NNs) (Feed forward or LSTM) to learn the pricing function of:

   (a) American Put/Call options, and

   (b) Barrier (down-and-out) Call options: this is a path-dependent option whose payoff depends on the the terminal price of the underlying asset and on whether the asset hits a specified barrier level during the life of the contract. In a down-and-out call option, the contract becomes worthless if underlying asset price ever falls below a barrier B prior to maturity T; otherwise it behaves like a European call option.

   The ground truth prices should be generated via **time-consuming** numerical methods such as:

   (a) Binomial tree for American options

   (b) Monte Carlo simulation with the Heston model for Barrier options

[1] However, you may use these in combination with other new indicators

Task Overview: Generate a synthetic dataset of option prices by sampling parameter combinations (strike, maturity, volatilities, etc.). More precisely: Input parameters to vary (you should choose appropriate ranges): Strike price K; Time to maturity T; Risk-free rate r; Dividend yield q; Volatility σ (for binomial). Heston model parameters (for MC): ν_0, θ, κ, η, ρ

Steps:

- Generate $n = 500$ - 2000 random parameter combinations. (ideal range (5000 -10000) but it will take many hours (it's your choice)).
- Compute option prices using the numerical pricing method (Binomial tree or Monte Carlo).
- Store dataset as tuples (X, y) where X = input parameters and y = computed price.
- Train a machine learning model on the dataset to learn the pricing function.
- Evaluate and report on the prediction accuracy and computational speedup.

3. (10%) Propose and implement in your solution to 2. improvements to the Euler MC method, so that simulations go faster